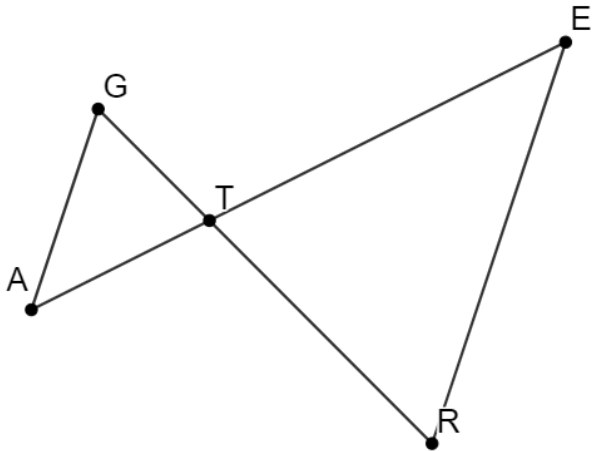




6.8.5 Wrap-Up: Ticket Out the Door

1. $\triangle GTA$ and $\triangle TER$ are shown.



Given: $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$

Prove: $\triangle GTA \sim \triangle RTE$

Complete the two-column proof.

Statement	Reason
1. $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$	1. Given
2. $\frac{TA}{TE} = \frac{2}{3}$, $\frac{TG}{RT} = \frac{2}{3}$	2.
3.	3. Transitive Property
4. $\angle GTA \cong \angle RTE$	4.
5. $\triangle GTA \sim \triangle RTE$	5.



Unit 6, Lesson 9: Proving Triangle Similarity – Part 2



Benchmark

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg.



Learning Target

- ☐ I can use triangle similarity to prove that triangles are similar.



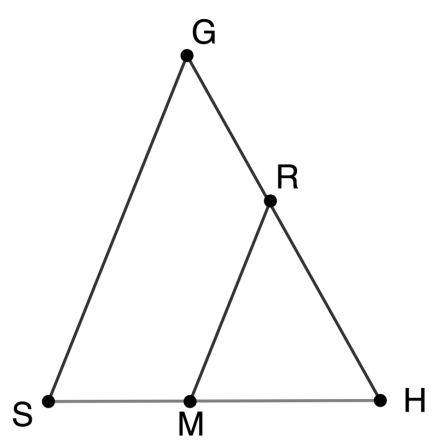
6.9.1 Warm-Up: Would You Rather?

1. In 2021, General Motors introduced the eVTOL air-taxi and the PAV (personal autonomous vehicle), a self-driving car. Both are concepts with no known timeline for production.
 - a. Do you think a drone-like personal vehicle like the eVTOL air-taxi will be reality in your lifetime?
 - b. Which would you rather have as your vehicle, an air-taxi or a self-driving car, in the future?
 - c. Explain your choice.



6.9.2 Activity: Proving Triangle Similarity

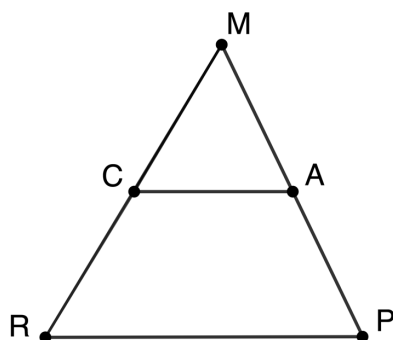
With your partner, sort through the statement and reason cards. Then, write each proof in the tables provided.

Proof A	
Given: $\overline{GS} \parallel \overline{RM}$ Prove: $\triangle RHM \sim \triangle GHS$ 	
Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.

Proof B

Given: $MC = \frac{2}{5}MR$ and
 $MA = \frac{2}{5}MP$

Prove: $\triangle MCA \sim \triangle MRP$

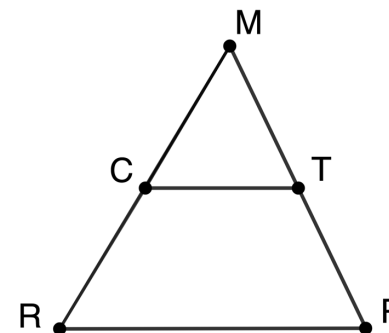


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Proof C

Given: $\angle RPT$ and $\angle CTP$ are supplementary

Prove: $\triangle MCT \sim \triangle MRP$

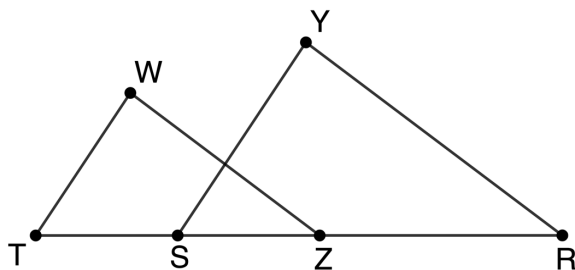


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.

Proof D

Given: $\overline{WT} \parallel \overline{YS}$,
 $\overline{WZ} \parallel \overline{YR}$

Prove:
 $\triangle WZT \sim \triangle YRS$

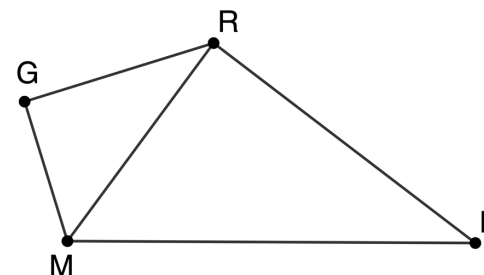


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Proof E

Given: $GM = \frac{3}{5}RM$,
 $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$

Prove:
 $\triangle MGR \sim \triangle MRB$

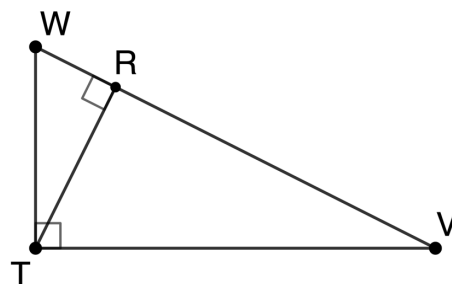


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.

Proof F

Given: $\overline{WT} \perp \overline{VT}$ and
 $\overline{WV} \perp \overline{RT}$

Prove: $\triangle RTW \sim \triangle TVW$

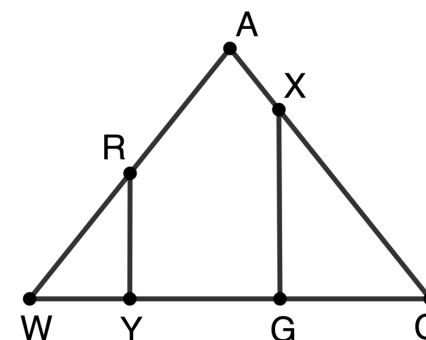


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

Proof G

Given: $\overline{WA} \cong \overline{CA}$,
 $\overline{RY} \perp \overline{WC}$, $\overline{XG} \perp \overline{WC}$

Prove: $\triangle RYW \sim \triangle XGC$



Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.



6.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about proving triangle similarity.

10.	10.
11.	11.
12.	12.
13.	13.
14.	14.